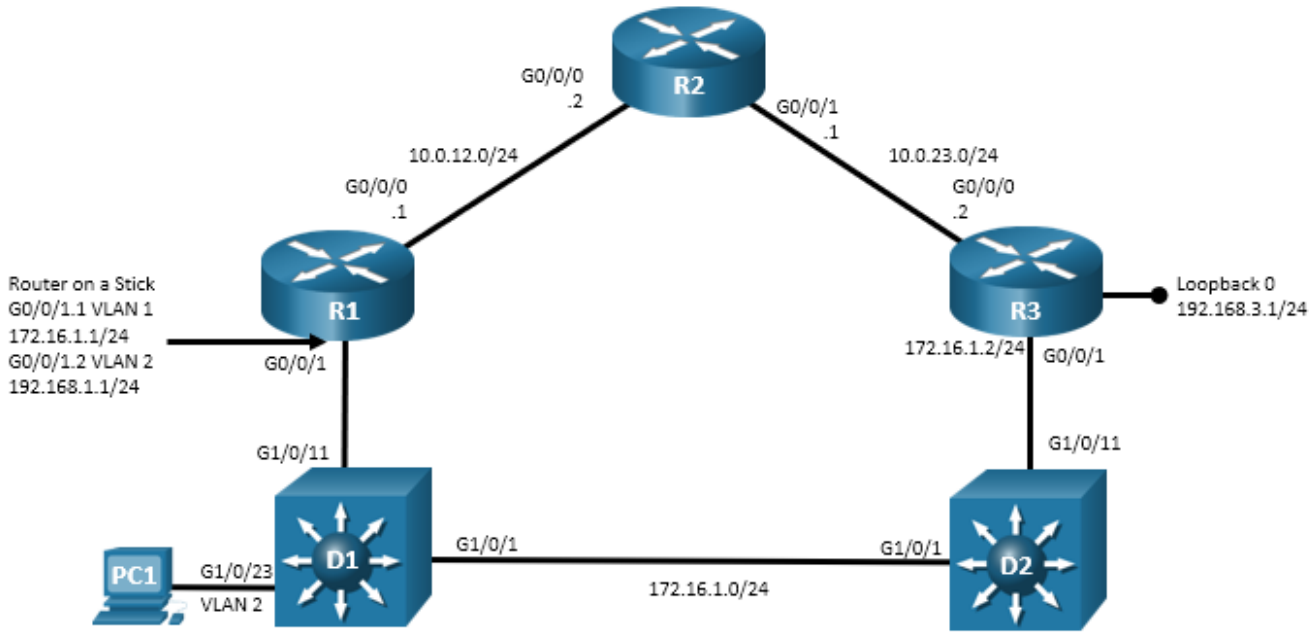
Lab - Troubleshoot EIGRP for IPv4

Topology



Addressing Table

Device	Interface	IP Address	Subnet Mask
R1	G0/0/0	10.0.12.1	255.255.255.0
	G0/0/1.1	172.16.1.1	255.255.255.0
	G0/0/1.2	192.168.1.1	255.255.255.0
R2	G0/0/0	10.0.12.2	255.255.255.0
	G0/0/1	10.0.23.1	255.255.255.0
R3	G0/0/0	10.0.23.2	255.255.255.0
	G0/0/1	172.16.1.2	255.255.255.0
	Loopback 0	192.168.3.1	255.255.255.0
PC1	NIC	DHCP	

Objectives

Troubleshoot network issues related to the configuration and operation of EIGRP for IPv4.

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Background / Scenario

In this topology, routers R1, R2, and R3 are EIGRP neighbors. Switches D1 and D2 provide connectivity between R1 and R3. Router R1 is configured for inter-VLAN routing and DHCP to provide support for PC 1. You will be loading configurations with intentional errors onto the network. Your tasks are to FIND the error(s), document your findings and the command(s) or method(s) used to fix them. FIX the issue(s) presented here and then test the network to ensure both of the following conditions are met:

1) the complaint received in the ticket is resolved
2) full reachability is restored

Note: The routers used with CCNP hands-on labs are Cisco 8200L with Cisco IOS XE Release 17.12.5 (universalk9 image). The switches used in the labs are Cisco Catalyst 9300L with Cisco IOS XE Release 17.12.5 (universalk9 image). Other routers, switches, and Cisco IOS versions can be used. Depending on the model and Cisco IOS version, the commands available and the output produced might vary from what is shown in the labs. Refer to the Router Interface Summary Table at the end of the lab for the correct interface identifiers.

Note: Make sure that the switches have been erased and have no startup configurations. If you are unsure, contact your instructor.

Required Resources

- 3 Routers (Cisco 8200L with Cisco IOS XE Release 17.12.5c universal image or comparable)
- 2 Switches (Cisco 9300L with Cisco IOS XE Release 17.12.5 universal image or comparable)
- 1 PC (Choice of operating system with terminal emulation program installed)
- Console cables to configure the Cisco IOS devices via the console ports
- Ethernet cables as shown in the topology

Instructions

Part 1: Trouble Ticket 4.1.1.1

Scenario:

During a planned capacity upgrade, R2 was added between R1 and R3 to provide additional bandwidth and a more resilient path between the LANs. After the change window, monitoring alerted that route exchange across the path was incomplete and that end users on the R1 side reported sporadic access to services behind R3. Interfaces show as operational, but the expected EIGRP neighbor relationships are not present across all links. Investigate and restore normal routing so that end-to-end reachability is consistent.

Be sure you are using devices that have been connected and prepared by your instructor. Use the commands listed below to load the configuration files for this trouble ticket:

Device	Command
R1	`copy flash:/enarsi/4.1.1.1-r1-config.txt running-config`
R2	`copy flash:/enarsi/4.1.1.1-r2-config.txt running-config`
R3	`copy flash:/enarsi/4.1.1.1-r3-config.txt running-config`
D1	`copy flash:/enarsi/4.1.1.1-d1-config.txt running-config`
D2	`copy flash:/enarsi/4.1.1.1-d2-config.txt running-config`

- PC 1 should be configured for and receive an address from an IPv4 DHCP server.

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- Passwords on all devices are **cisco12345**. If a username is required, use **admin**.
- After you have fixed the ticket, change the MOTD on EACH DEVICE using the following command:

  banner motd # This is $(hostname) FIXED from ticket <ticket number> #
- Then save the configuration by issuing the **wri** command (on each device).
- Inform your instructor that you are ready for the next ticket.
- After the instructor approves your solution for this ticket, issue the **reset.now** privileged EXEC command on each device. This script will clear your configurations and reload the devices.

Part 2: Trouble Ticket 4.1.1.2

Scenario:

An external consultant performed a security hardening of the routing domain with the goal of tightening route control and adding authentication where appropriate. Following the maintenance, teams noticed that networks behind R1 were no longer being learned across R2, resulting in loss of access from the R3 side to resources on R1's LANs. Physical connectivity and interface status look healthy. Review the EIGRP configuration and policies so that legitimate routes are exchanged again without weakening the intended security posture.

Use the commands listed below to load the configuration files for this trouble ticket:

Device	Command
R1	`copy flash:/enarsi/4.1.1.2-r1-config.txt running-config`
R2	`copy flash:/enarsi/4.1.1.2-r2-config.txt running-config`
R3	`copy flash:/enarsi/4.1.1.2-r3-config.txt running-config`
D1	`copy flash:/enarsi/4.1.1.2-d1-config.txt running-config`
D2	`copy flash:/enarsi/4.1.1.2-d2-config.txt running-config`

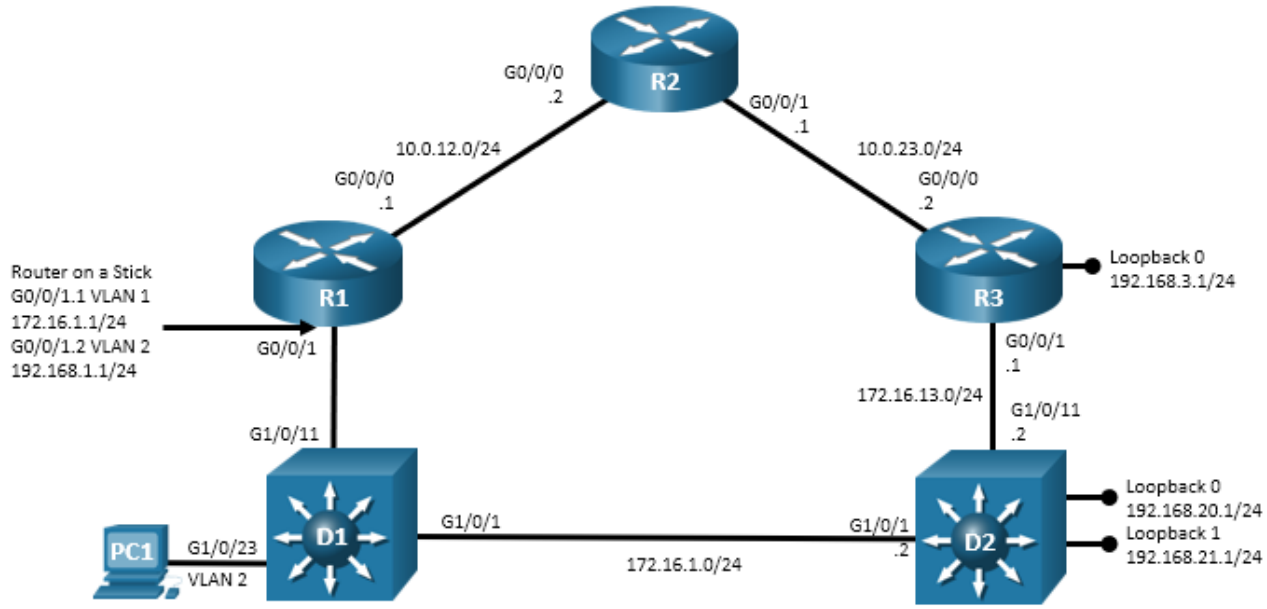
- PC1 should be configured for and receive an address from an IPv4 DHCP server.
- Passwords on all devices are **cisco12345**. If a username is required, use **admin**.
- After you have fixed the ticket, change the MOTD on EACH DEVICE using the following command:

  banner motd # This is $(hostname) FIXED from ticket <ticket number> #
- Then save the configuration by issuing the **wri** command (on each device).
- Inform your instructor that you are ready for the next ticket.
- After the instructor approves your solution for this ticket, issue the **reset.now** privileged EXEC command. This script will clear your configurations and reload the devices.

Part 3: Trouble Ticket 4.1.1.3

Topology Update:

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Addressing Table Update:

Device	Interface	IP Address	Subnet Mask
R1	G0/0/0	10.0.12.1	255.255.255.0
	G0/0/1.1	172.16.1.1	255.255.255.0
	G0/0/1.2	192.168.1.1	255.255.255.0
R2	G0/0/0	10.0.12.2	255.255.255.0
	G0/0/1	10.0.23.1	255.255.255.0
R3	G0/0/0	10.0.23.2	255.255.255.0
	G0/0/1	172.16.1.2	255.255.255.0
	Loopback 0	192.168.3.1	255.255.255.0
D2	G1/0/1	172.16.1.2	255.255.255.0
	G1/0/11	172.16.13.2	255.255.255.0
	Loopback 0	192.168.20.1	255.255.255.0
	Loopback 1	192.168.21.1	255.255.255.0
PC1	NIC	DHCP	

Scenario:

As part of a distribution-layer refresh, D2 was converted to Layer 3 and connected to the network using routed links to R1 and R3. D2 is expected to participate in the existing EIGRP domain, form neighborships on both uplinks, and advertise its new internal networks. After cutover, D2 does not appear as an EIGRP neighbor on either peer, and its local routes are not present elsewhere in the domain. Validate and align D2's routing parameters with the domain so that adjacencies form and the new networks are reachable.

Use the commands listed below to load the configuration files for this trouble ticket:

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Device	Command
R1	`copy flash:/enarsi/4.1.1.3-r1-config.txt running-config`
R2	`copy flash:/enarsi/4.1.1.3-r2-config.txt running-config`
R3	`copy flash:/enarsi/4.1.1.3-r3-config.txt running-config`
D1	`copy flash:/enarsi/4.1.1.3-d1-config.txt running-config`
D2	`copy flash:/enarsi/4.1.1.3-d2-config.txt running-config`

- PC1 should be configured for and receive an address from an IPv4 DHCP server.
- Passwords on all devices are **cisco12345**. If a username is required, use **admin**.
- After you have fixed the ticket, change the MOTD on EACH DEVICE using the following command:
  banner motd # This is $(hostname) FIXED from ticket <ticket number> #
- Then save the configuration by issuing the **wri** command (on each device).
- Inform your instructor that you are ready for the next ticket.
- After the instructor approves your solution for this ticket, issue the **reset.now** privileged EXEC command. This script will clear your configurations and reload the devices.

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